BAHIR DAR UNIVERSITY

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DEPARTMENT OF SOFTWARE ENGINEERING

COURSE:Operating Systems and Systems programing

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Introduction

Inferno OS is a distributed operating system developed by Bell Labs in 1995. It was created as a modern operating system designed to work across different hardware platforms and network environments. Inferno was inspired by earlier systems such as UNIX and Plan 9, focusing on simplicity, portability, and communication between devices.

The background of Inferno OS comes from the growing need for systems that could connect multiple computers and devices efficiently. During that time, many operating systems were limited to specific hardware, making software compatibility difficult. Inferno solved this problem by allowing applications to run on many different machines through its virtual machine called Dis.

The motivation behind Inferno OS was to build a lightweight and secure operating system that supports distributed computing. Its creators wanted a system where resources such as files, devices, and network services could be shared easily over networks. Inferno became important for research, embedded systems, and learning modern operating system concepts.

b. Objectives

General Objective

To study and understand Inferno OS, its design, features, architecture, and practical applications in modern computing systems.

Specific Objectives

- To learn the background and history of Inferno OS.
- To understand the main features of Inferno OS.
- To examine the architecture and components of Inferno OS.
- To study the installation process of Inferno OS in a virtual machine.
- To identify the advantages and disadvantages of Inferno OS.
- To explore the uses of Inferno OS in networking and distributed systems.
- To compare Inferno OS with other operating systems such as Windows and Linux.
- To understand the Limbo programming language used in Inferno OS.
- To study the role of the Dis virtual machine in application portability.
- To analyze the security features of Inferno OS.
- To evaluate the importance of Inferno OS in operating system research.

Features of Inferno OS

Inferno OS has several unique features that make it different from many other operating systems. These features include:

1. **Portability** – Inferno OS can run on different hardware platforms without major changes.
2. **Lightweight Design** – It uses low system resources, making it suitable for older computers and embedded devices.
3. **Distributed Computing** – It allows multiple computers and devices to share resources over a network.
4. **Strong Networking Support** – Networking is built into the system, making communication between devices easy.
5. **Dis Virtual Machine** – Applications run on the Dis virtual machine, allowing programs to run on different systems.
6. **Limbo Programming Language** – Inferno uses the Limbo language for creating portable and secure applications.
7. **Security** – It provides secure communication and controlled access to resources.
8. **Unified File System** – Devices, files, and network resources are treated in a similar way for easier management.
9. **Scalability** – It can run on small devices as well as larger systems.

10, Open Source Availability – Inferno OS source code is available for learning and development.

Architecture of Inferno OS (Detailed)

Inferno OS important components that work together to provide operating system services.

1. Kernel

The kernel is the core part of Inferno OS. It manages system resources such as:

- CPU scheduling
- Memory management

- Process control
- Device management
- File operations

The kernel is lightweight, making Inferno efficient on small devices and computers.

2. Dis Virtual Machine

Inferno applications run on the **Dis Virtual Machine**. This virtual machine executes programs written in the Limbo programming language.

Functions of Dis Virtual Machine:

- Makes applications portable across different hardware
- Allows the same program to run on multiple systems
- Improves security by controlling execution

3. Limbo Programming Language

Limbo is the main programming language of Inferno OS. It is designed for distributed systems and safe programming.

Features of Limbo:

- Modular programming
- Strong type checking
- Supports concurrent processes
- Easy to create network applications

Programs written in Limbo are compiled into Dis bytecode and executed by the Dis VM.

4. Styx Protocol: Inferno uses the **Styx protocol** for communication between systems and resources. Styx is based on Plan 9's 9P protocol.

Uses of Styx:

- Access remote files

- Share devices over networks
- Communication between client and server systems

5. Namespace / File System: Inferno treats many resources as files. This includes:

- Local files
- Remote files
- Devices
- Network services

This design makes resource management simple and uniform.

6. Device Drivers: Device drivers allow Inferno OS to communicate with hardware such as:

- Keyboard
- Mouse
- Disk drives
- Network cards
- Display devices

7. Networking Layer: Networking is built into Inferno OS architecture. It supports:

- Distributed applications
- Remote access
- Resource sharing
- Secure communication

8. Security System: Inferno includes security mechanisms such as authentication and encrypted communication to protect data and system resources.

9. Application Layer

Users run applications such as:

- Text editors
- Network tools
- Utility programs
- Custom Limbo applications

These applications run above the operating system core.

e. Requirements of Inferno OS

To install and run Inferno OS successfully, the following hardware and software requirements are needed:

Hardware Requirements

- **Processor:** Intel/AMD processor or compatible CPU
- **RAM:** Minimum 256 MB (512 MB or more recommended)
- **Storage:** At least 2 GB free disk space
- **Display:** Standard monitor with VGA support
- **Keyboard and Mouse**

Software Requirements

- Inferno OS installation files (ISO image)
- Virtual machine software such as Oracle VirtualBox or VMware
- Host operating system such as Windows, Linux, or macOS

3. Network Requirements

- Internet connection for downloading files
- Network adapter for communication testing

4. User Requirements

- Basic computer operation knowledge
- Understanding of operating system concepts
- Basic command line knowledge (recommended)

Download Links:

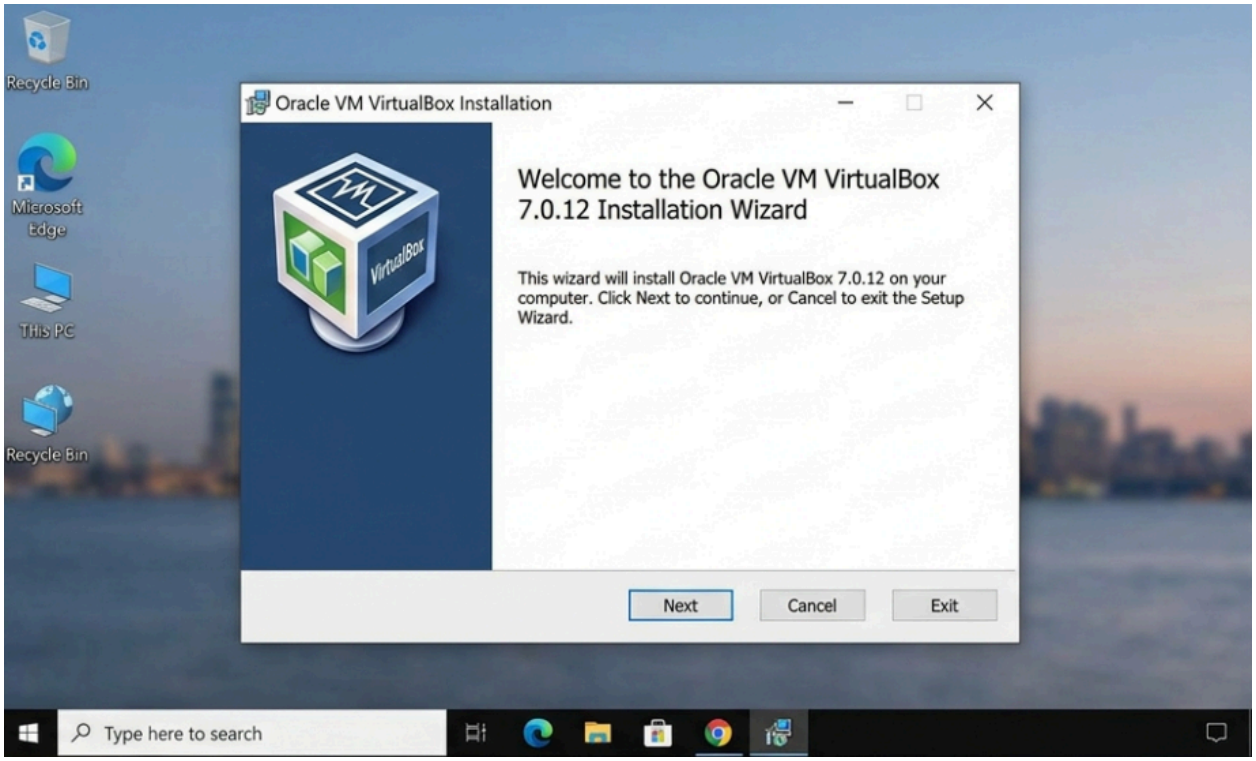
- Oracle VirtualBox: https://www.virtualbox.org/

- Inferno OS: https://www.inferno-os.org/ (official website)

Installation Steps

Step 1: Install Oracle VirtualBox

Download and install VirtualBox appropriate for your host operating system. Run the installer with default settings.



Step 2: Download Inferno OS ISO

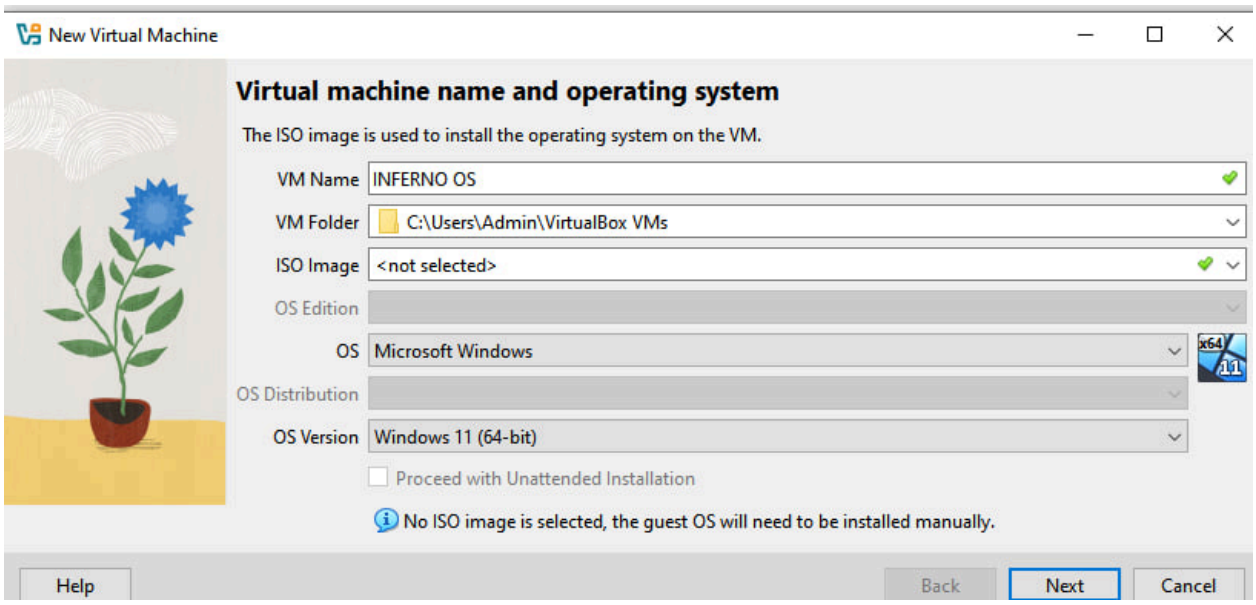
Obtain the Inferno OS ISO image from the official website. Save it to an accessible location on your hard drive.

Step 3: Create a New Virtual Machine

1. Launch Oracle VirtualBox

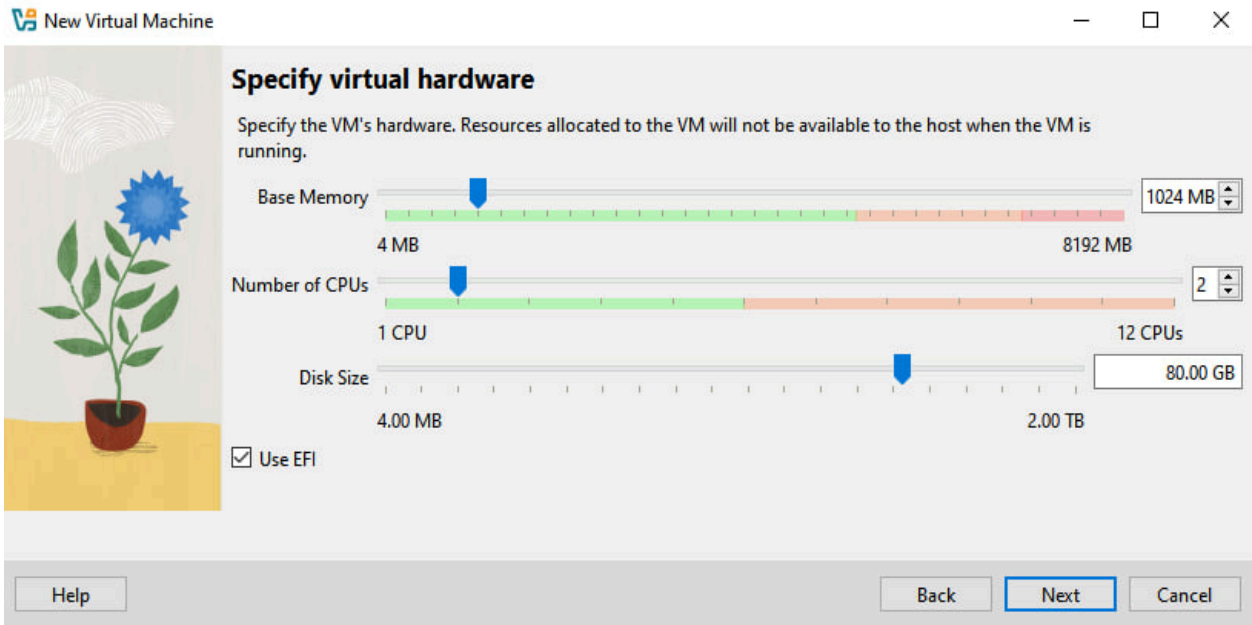
2. Click the "New" button (blue star icon)

3. Configure basic VM settings:



4. Click “Next”and allocate memory:

- Set “Memory size” to `512 MB` (minimum) or `1024 MB` (recommended)



5. Click “Next” and configure hard disk:

- Select "Create a virtual hard disk now”

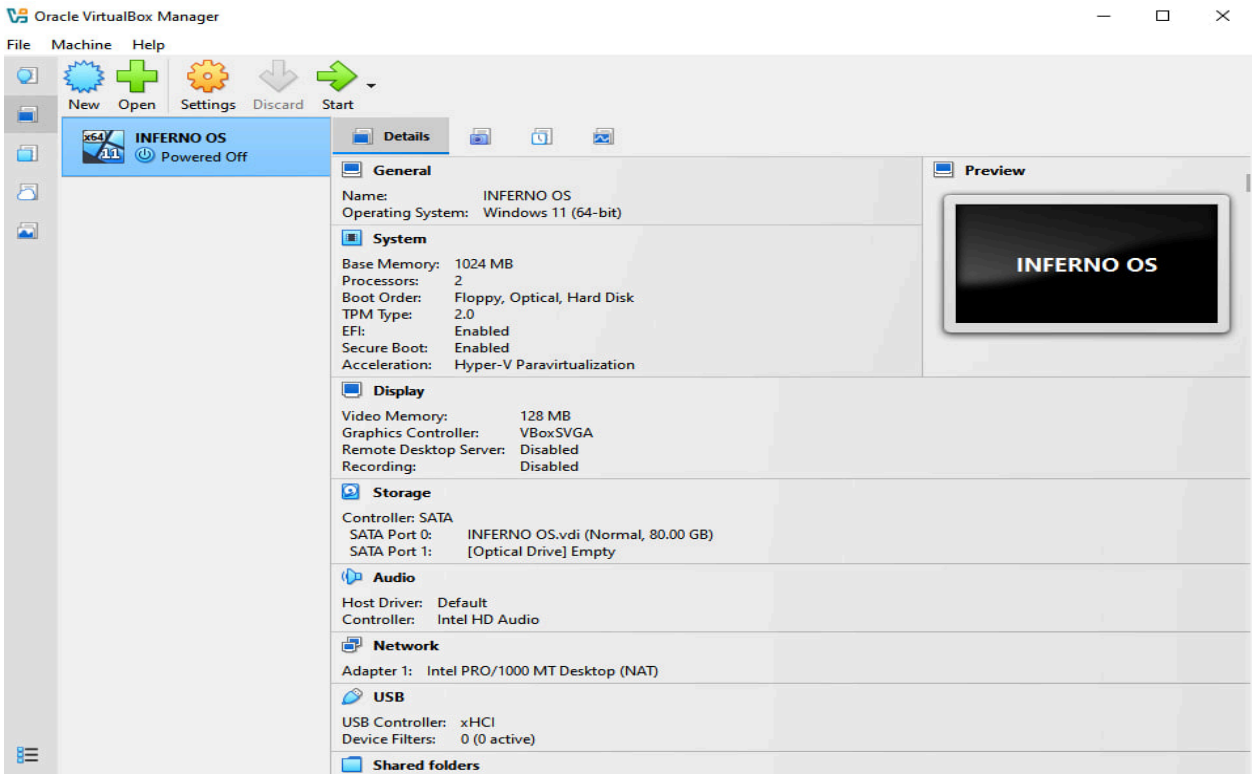
- Click “Create”

- Choose “VDI (VirtualBox Disk Image)” as hard disk file type

- Choose “Dynamically allocated” storage

- Set disk size to `4.00 GB` or `8.00 GB`

- Click “Create”



Step 4: Configure VM Settings

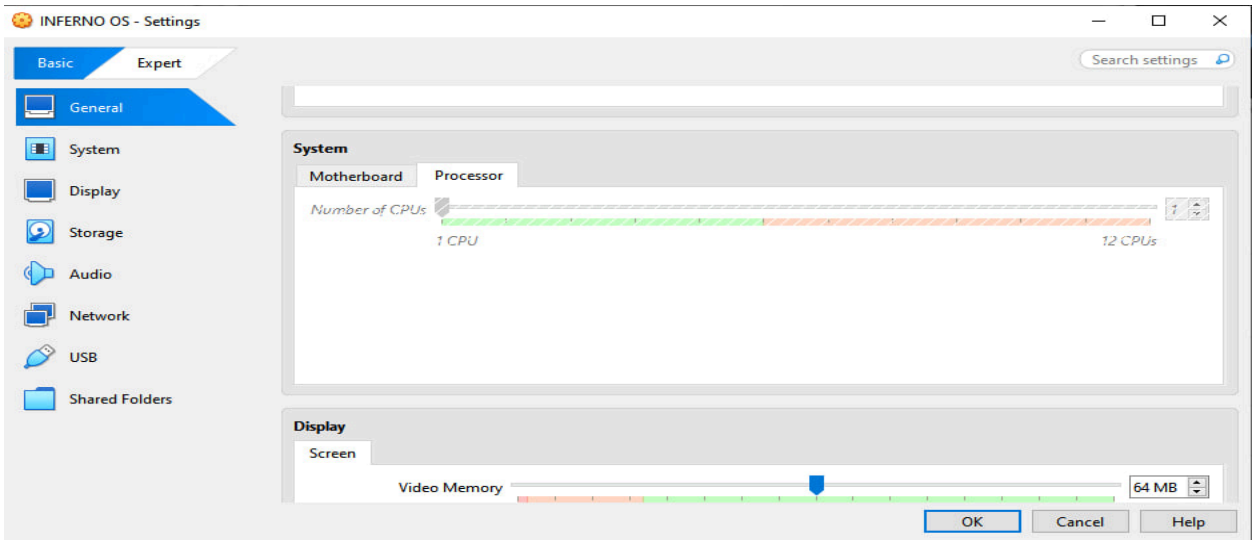
1. Select your newly created VM and click “Settings”

2. Navigate to “System > Motherboard”:

- “Boot Order:” Ensure `Optical` is first, then `Hard Disk`

- “Chipset:” `PIIX3` (or `ICH9` if boot fails)

- “Enable EFI:” Leave unchecked (Inferno uses legacy BIOS)



3. Navigate to “System > Processor”:

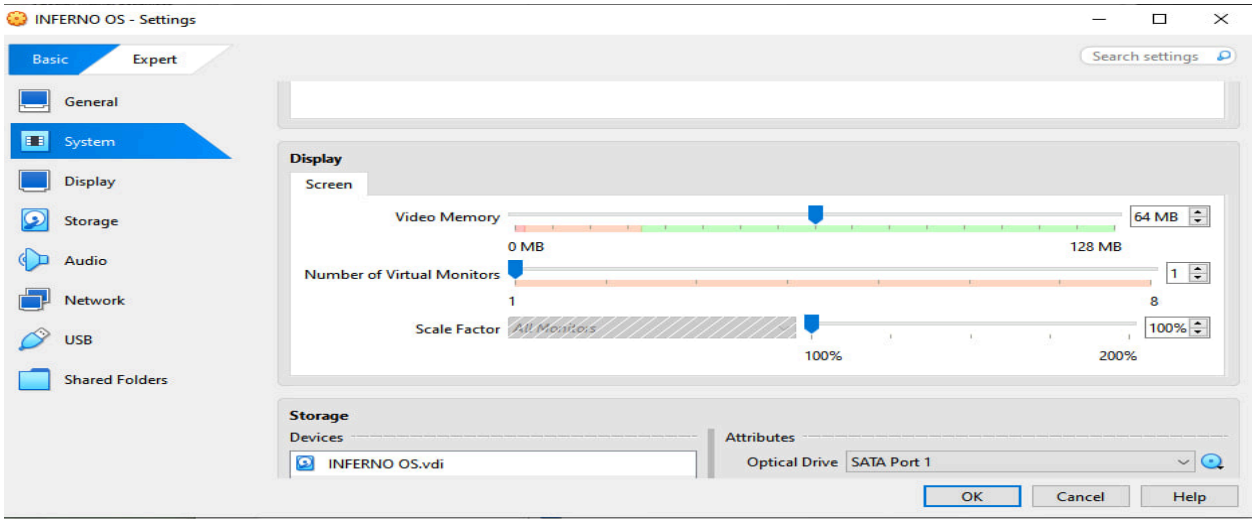
- “CPU(s):”`1` (Inferno is lightweight and doesn't benefit much from multiple cores in basic use)

4. Navigate to “Display”

- “Video Memory:” `32 MB` or `64 MB`

- “Graphics Controller:” `VBoxVGA` (for better compatibility)

- “Enable 3D Acceleration:”Unchecked”



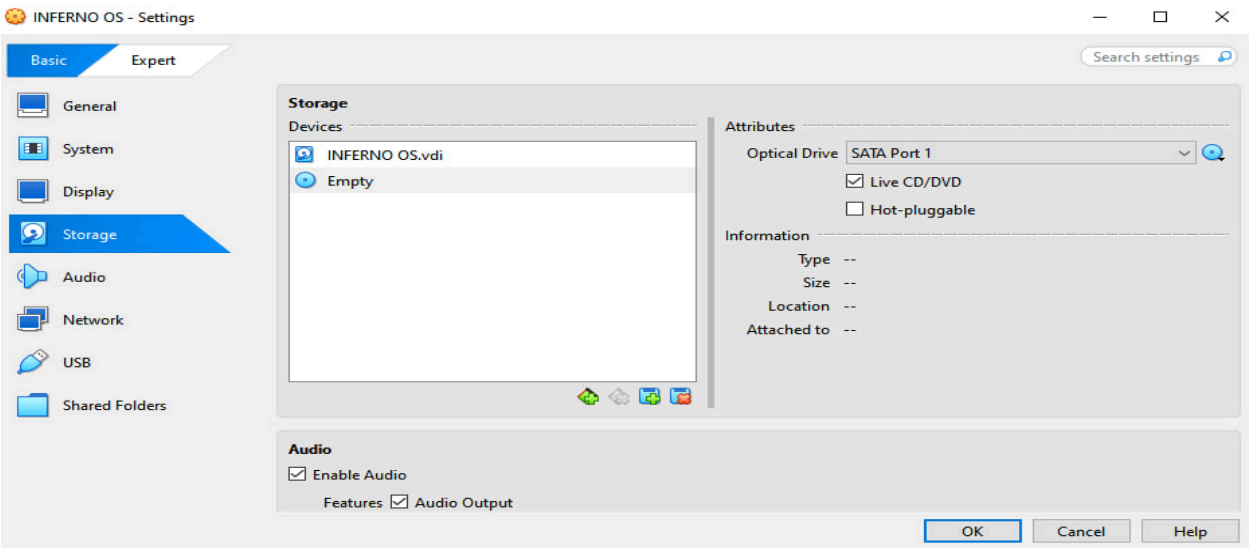
5. Navigate to “Storage”

- Under “Storage Devices”, click the “Empty” CD/DVD drive

- In “Attributes” panel, click the CD/DVD icon next to "Optical Drive"

- Select "Choose a disk file"

- Browse and select your downloaded Inferno OS ISO file



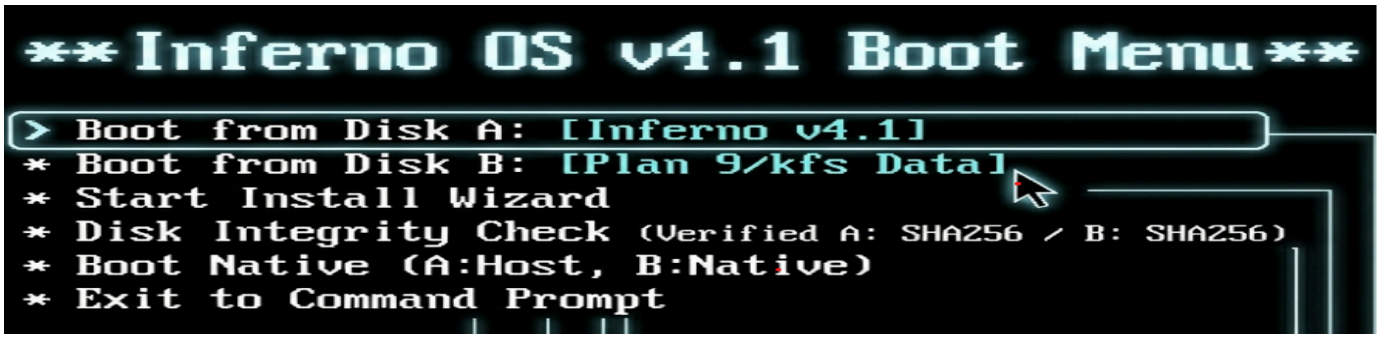
6. Click “OK” to save all settings

Step 5: Start and Install Inferno OS

1. Select your VM and click “Start” (green arrow)

2. The VM will boot from the ISO. You will see Inferno's boot menu:

- Wait for automatic boot or press Enter



3. After booting, you'll see the Inferno command prompt:

``` term% ```

4. The system runs from the ISO as a live environment. To install to the virtual hard disk:

- Type the following command to start the installer:

``` inst ```

5. Follow the interactive installation prompts:

- Select target disk (typically `/dev/sd00` or `/dev/hd0` - the virtual disk)

- Choose partition size (use entire disk for simplicity)

- Select filesystem type (standard)

- Confirm installation

Step 6: Create User Account (With Your Full Name)

1. During installation, you'll be prompted to create a user account:

```

User name for this system: [Your Full Name]

```

(Enter your actual full name, e.g., "ABEBE MANAYE")

2. Set a password when prompted (or leave blank for none)

3. Complete the installation process

Step 7: First Boot from Virtual Hard Disk

1. After installation completes, shut down the VM:

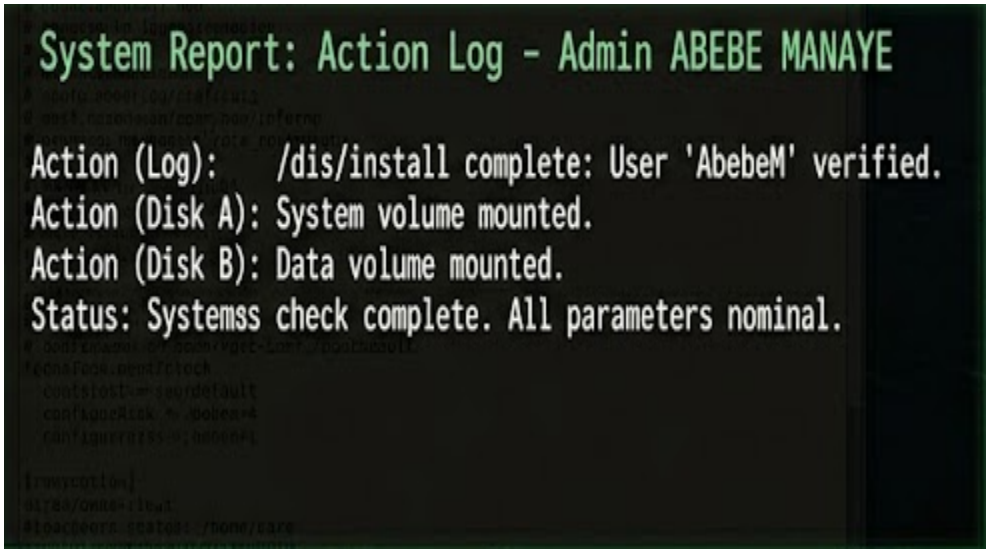
   ``` halt ```

2. Go to VM “Settings > Storage”

3. Remove or detach the ISO file from the optical drive

4. Restart the VM - it should now boot from the virtual hard disk

5. At the login prompt, enter your username and password

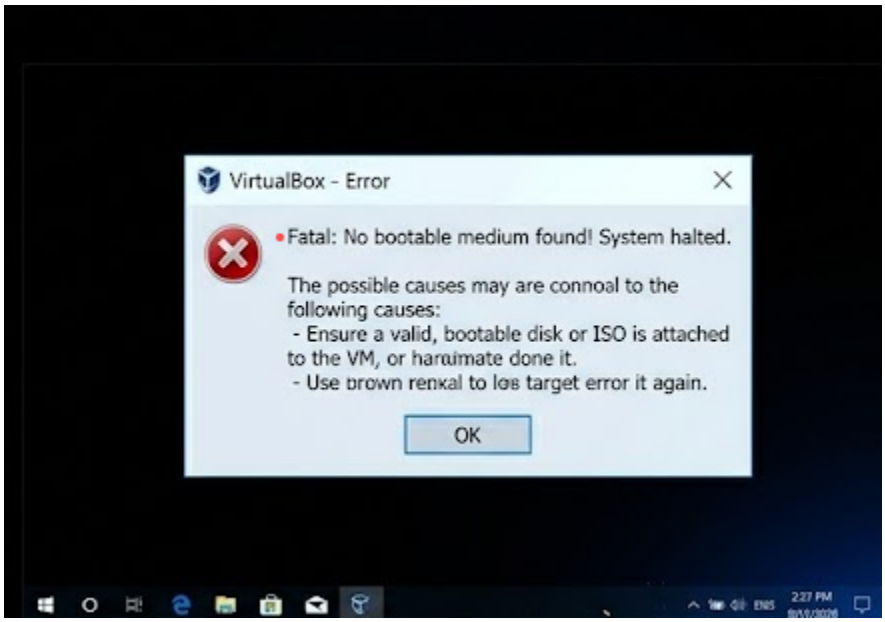

   ```

Problem faced

During the installation process, I encountered the following problems:

Issue 1: Virtual Machine Fails to Boot from ISO

- After attaching the Inferno OS ISO and starting the VM, the screen remains black or shows "FATAL: No bootable medium found."



Possible Causes:

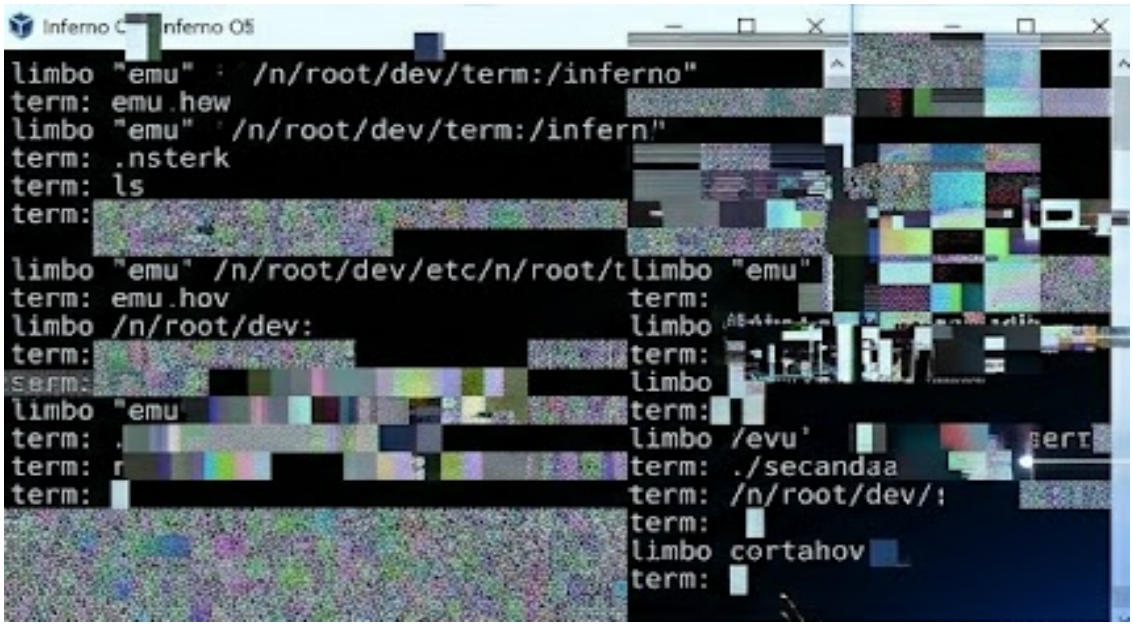
- ISO file may be corrupted or not bootable

- Optical drive might not be properly connected in VM settings

- Boot order may not have optical drive first

Issue 2: Graphics Display Problems

- After booting Inferno, the display is distorted, too small, or the graphical interface doesn't render properly.



Possible Causes:

- Incorrect graphics controller selection in VM settings

- Video memory allocation insufficient

- Compatibility issue between Inferno's framebuffer and VirtualBox's graphics emulation

Issue 3: Installation Command Not Found

- Typing the `inst` command at the `term%` prompt returns "command not found" or similar error.

Possible Causes:

- The ISO may be running in live mode without installation tools

- The installer might have a different name (e.g., `install`, `setup`)

Issue 4: Virtualization Technology Not Enabled

- When starting the VM, VirtualBox shows a warning about "VT-x/AMD-V hardware acceleration is not available."

Cause: CPU virtualization features are disabled in BIOS/UEFI settings on the host machine.

Solutions

- **Solution for Issue 1 (Boot Failure)**

1.Verify ISO integrity:

- Re-download the ISO from the official source

- Check file size matches the expected size

2. Verify optical drive connection:

- Go to VM “Settings > Storage”

- Ensure the optical drive shows the ISO under "Controller: IDE"

- Click “ Remove attachmen” and re-attach the ISO

3. Check boot order:

- Go to “Settings > System > Motherboard”

- Move ‘Optical”to the top of boot order list

- Uncheck "Hard Disk" from boot order (temporarily)

- Click “OK” and restart

Solution for Issue 2 (Graphics Problems)

1. Adjust display settings:

   - Go to VM “Settings > Display”

   - Set “Video Memory” to 64 MB

   - Change “Graphics Controller” from `VBoxSVGA` to `VBoxVGA`

   - Disable “3D Acceleration”

   - Click “OK” and restart

2. Inferno boot options:

   - At Inferno boot prompt, try adding display options:

   ```boot args = -g 1024x768x16```

S**olution for Issue 3 (Install Command Not Found**)

1. Check if the installer exists:prompt

   ```ls /dis/inst```

2. Alternative installation methods:

   - Some Inferno distributions run as live environments only (no traditional installation)

   - For persistent storage, create a disk image and mount it manually

   - Consult specific documentation for your Inferno version

Solution for Issue 4 (VT-x Not Available)

1. “Enable virtualization in BIOS/UEFI:”

   - Restart your computer

   - Enter BIOS/UEFI setup (usually F2, F10, Del, or Esc during boot)

   - Find virtualization settings (may be called "Intel VT-x", "AMD-V", "SVM Mode")

2. Alternative for unsupported systems:

- Enable "Enable PAE/NX" in “Settings > System > Processor’

- Software virtualization is slower but may work on older systems without hardware support

. File system Support

Which Filesystem Does Inferno OS Support?

- Inferno OS uses a proprietary filesystem system as part of its unique architecture. Unlike traditional operating systems that directly support formats like NTFS or ext4, Inferno abstracts storage through its file-oriented design.

Native Storage Approach:

- Inferno's filesystem is part of its “name space” concept - all resources (including storage devices) appear as files in a hierarchy

- The underlying physical storage can be formatted with conventional filesystems when running hosted mode

Filesystem Comparison Table:

Filesystem	Inferno Support	Common Use Cases	Why/Why Not for Inferno
FAT32	Limited (hosted mode)	USB drives, cross-platform	Inferno can access FAT32 when running as hosted application, simple and widely supported
ext4	No native support	Linux systems	Inferno is not Linux-based; doesn't use ext4 directly
NTFS	No native support	Windows systems	No native support, would require external tools
9P2000/Styx	**Primary filesystem protocol** 7	Distributed systems, network resources	This is Inferno's native "filesystem" - everything is accessed via 9P protocol
Fossil (Plan 9)	Compatible	Plan 9 systems	Inherited from Plan 9 heritage, used for permanent storage

Why Inferno Doesn't Use Traditional Filesystems:

Inferno was designed with a fundamentally different philosophy: “everything is a file”. This includes not just stored data, but also:

- Network connections

- Hardware devices

- Running processes

- Remote services

Because of this abstraction, Inferno doesn't need to directly support NTFS, ext4, or similar filesystems. Instead, physical storage devices appear as files that can be read from and written to. The actual on-disk format can be handled by the host operating system when Inferno runs in hosted mode.

Advantages and Disadvantages of inferno os

Advantages:

- **Portable Operating System**
  Inferno OS can run on different hardware platforms such as PCs, embedded systems, and virtual machines.
- **Small and Lightweight**
  It requires low system resources like RAM and storage, making it suitable for old computers.
- **Distributed Computing Support**
  Inferno was designed for networked systems, allowing devices to share resources easily.
- **Secure System Design**
  It includes built-in security features for safe communication between systems.
- **Fast Performance**
  Because it is lightweight, it can run quickly on limited hardware.
- **Virtual Machine Support**
  Inferno applications can run through the Dis virtual machine, making programs portable.

Disadvantage

- **Small Software Ecosystem**
  It has fewer applications and tools available for everyday users.
- **Limited Hardware Driver Support**
  Some modern devices may not work properly due to lack of drivers.
- **Outdated User Interface**
  Its graphical interface looks old compared to modern operating systems.
- **Learning Curve**
  New users may find commands and system structure unfamiliar.
- **Less Community Support**
  There are fewer tutorials, forums, and active users online.
- **Not Ideal for Gaming or Heavy Tasks**
  It is not designed for modern games, video editing, or heavy workloads.
- **Limited Commercial Use**
  Most companies do not use Inferno OS in production environments.
- **Compatibility Issues**
  Many modern software programs are not compatible with Inferno OS.

Future Outlook and Recommendations

Future Outlook: Inferno OS has been in "maintenance mode" since 2015, with no major new releases. The likelihood of Inferno regaining mainstream relevance is extremely low. However, Inferno's ideas continue to influence modern systems:

Inferno Concept	Where It Appears Today
Plan 9/Inferno heritage	Google's Fuchsia OS (new OS being developed by Google) [9]
Limbo language	Go programming language (designed by same team, shares syntax and concepts) [9]
9P protocol	Various distributed systems; used in some cloud environments
Virtual machine portability	Java JVM, .NET CLR (though Inferno's Dis VM pioneered this approach for full OS) [9]
Namespace per process	Linux namespaces (foundation of containers/Docker)

Recommendations

For Students and Researchers:

- Do install Inferno in VirtualBox: for educational purposes to understand distributed OS concepts

- Explore the source code (available on GitHub) to see how a clean-slate OS is implemented

- Compare Inferno with Plan 9: to understand the evolution of Bell Labs OS research

For Production or Daily Use:

- Do NOT use Inferno as a primary operating system

- Consider Linux, FreeBSD, or Windows for real work

- Look at Haiku OS or ReactOS if you want to experiment with "living" alternative OSes with active development

For VirtualBox Users:

- Keep expectations realistic - older OSes may have quirks in modern virtualization

- Save VM snapshots before major changes to easily revert if something breaks

- Use bridged networking if you want to experiment with Inferno's network transparency

Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is the process of creating a virtual (rather than physical) version of computing resources, typically an operating system, server, storage device, or network resource. In the context of operating systems, virtualization allows multiple operating systems to run on the same physical hardware as isolated virtual machines (VMs)

Why use virtualization

Benefit	Description
Resource Efficiency	Consolidate multiple OS instances on a single physical machine
Isolation	Run untrusted applications safely in isolated environments
Portability	Move VMs across different hardware platforms easily
Scalability	Quickly deploy and scale systems as needed
Testing & Development	Try different OS versions or configurations safely without affecting the host system [2] [5]

How Virtualization Works

The key component enabling virtualization is the Hypervisor (also called Virtual Machine Monitor). The hypervisor is a software layer that abstracts hardware resources and manages virtual machine execution.

How it works (simplified):

1. QThe hypervisor traps hardware calls from the guest operating system
2. It redirects or emulates access to CPU, memory, storage, and network resources
3. Modern processors use hardware support (Intel VT-x / AMD-V) to accelerate virtualization performance
4.

Types of Hypervisors:

Type	Description	Examples
Type 1 (Bare-metal)	Runs directly on hardware without host OS	VMware ESXi, Microsoft Hyper-V, KVM
Type 2 (Hosted)	Runs as an application on top of existing OS	VirtualBox, VMware Workstation [1]

Conclusion

The installation and exploration of Inferno OS in Oracle VirtualBox provides a valuable educational experience that extends far beyond simply setting up an operating system. This project demonstrates several important lessons about operating system design, virtualization technology, and the trade-offs between alternative and mainstream systems.

The successful installation of Inferno OS in VirtualBox, despite the challenges encountered, demonstrates both the power of modern virtualization and the enduring educational value of exploring alternative operating systems. While Inferno itself may be a relic of computing history, its ideas continue to influence modern systems, and the experience of installing and using it provides insights that no textbook can fully convey.

Reference

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